MAR ATHANASIOS COLLEGE FOR ADVANCED STUDIES, TIRUVALLA
(MACFAST)

(Affiliated to Mahatma Gandhi University, Kottayam)

MACFAST (AUTONOMOUS)

MCACP 305 PYTHON PROGRAMMING FOR DATA SCIENCE

Department of Computer Applications

Mar Athanasios College for Advanced Studies, Tiruvalla

(MACFAST) AUTONOMOUS

Kerala, India, Pin-689101

www.macfast.org

www.macfast.ac.in

MAR ATHANASIOS COLLEGE FOR ADVANCED STUDIES, TIRUVALLA
(MACFAST)

(Affiliated to Mahatma Gandhi University, Kottayam)

MACFAST (AUTONOMOUS)

MCACP 305 PYTHON PROGRAMMING FOR DATA SCIENCE

Name	**:**	**JERIN GEORGE**
Register No	**:**	**253242210231**
Semester	**:**	**3**
Subject	**:**	**PYTHON PROGRAMMING FOR DATA SCIENCE**

Certified that this is a bonafide record of the practical work done by Mr. **JERIN GEORGE** *Reg.No:* **253242210231** *in the Department of Computer Applications during 2025-2027 at Mar Athanasios College for Advanced Studies, Tiruvalla ((MACFAST) Autonomous).*

Faculty in Charge **Head of the Department**

Examiners

1……

2…….

INDEX

SL.NO	DATE	NAME OF PROGRAM	PAGE NO
1	03/06/2026	Given number between a range	6
2	03/06/2026	Distance between 2 points using Pythagoras theorem	8
3	03/06/2026	Library Management System	10
4	10/06/2026	Fibonacci Series with in a limit	12
5	10/06/2026	Prime numbers within a range	14
6	10/06/2026	Display today’s date & current time	16
7	17/06/2026	Python List Operations	18
8	17/06/2026	Arithmetic Division Using Multiple Exception	20
9	24/06/2026	Exception handling using division by ZeroDivisionError	22
10	01/07/2026	Factorial of a number using function	24
11	06/07/2026	Arithmetic operations using function	26
12	06/07/2026	Vehicle Class with Maximum speed & Mileage	30
13	08/07/2026	Constructor & Destructor Demonstration	32
14	08/07/2026	Functions to find the longest line in a file	34
15	13/07/2026	Creating & importing a calculator Module	36

16	16/07/2026	SQLITE3 Operations	40
17	20/07/2026	Data preprocessing toolkit with Pandas and NumPy	44
18	17/08/2026	Data preprocessing toolkit with matplotlib	56
19	17/08/2026	Simple linear regression	60
20	07/09/2026	Classification based logistic regression	62
21	14/09/2026	DBSCAN using clustering	68
22	28/09/2026	Django Framework - Static Web Page	72
23	05/10/2026	Django Framework – Dynamic Web Page	80

OUTPUT

```
========================= RESTART: C:\Python3_jas\num.py =======
Range:1-100
Enter a number:24
The number  24 is in range
>>>
========================= RESTART: C:\Python3_jas\num.py =======
Range:1-100
Enter a number:110
The number is not in range
>>>
```

PROGRAM-1

GIVEN NUMBER BETWEEN A RANGE

AIM: Python program to Check whether a given number is between a range.

SOURCE CODE:

```
num = int(input("Enter a number: "))
lower = int(input("Enter lower limit: "))
upper = int(input("Enter upper limit: "))
if lower <= num <= upper:
  print("The number is within the range.")
else:
  print("The number is outside the range.")
```

RESULT: THE PROGRAM IS COMPILED SUCCESSFULLY AND THE OUTPUT IS VERIFIED.

OUTPUT

```
Enter x1: 2
Enter y1: 3
Enter x2: 4
Enter y2: 1
Distance = 2.8284271247461903
```

PROGRAM-2

DISTANCE BETWEEN 2 POINTS USING PYTHAGORAS THEOREM

AIM: Python program to compute the distance between two points using Pythagoras Theorem.

SOURCE CODE:

```
import math
x1 = float(input("Enter x1: "))
y1 = float(input("Enter y1: "))
x2 = float(input("Enter x2: "))
y2 = float(input("Enter y2: "))
distance = math.sqrt((x2 - x1)**2 + (y2 - y1)**2)
print("Distance =", distance)
```

RESULT: THE PROGRAM IS COMPILED SUCCESSFULLY AND THE OUTPUT IS VERIFIED.

OUTPUT

```
Enter the number of late days: 0
No fine.

=== RESTART: C:\Users\print\AppData\Local\Prog
Enter the number of late days: 4
Fine = Rs. 8

=== RESTART: C:\Users\print\AppData\Local\Prog
Enter the number of late days: 8
Fine = Rs. 24

=== RESTART: C:\Users\print\AppData\Local\Prog
Enter the number of late days: 26
Fine = Rs. 130
```

PROGRAM-3

LIBRARY MANAGEMENT SYSTEM

AIM: Python program to accept the number of days a member is late in returning a book and display the fine.

SOURCE CODE:

```
days = int(input("Enter the number of late days: "))
if days == 0:
  print("No fine.")
elif days <= 5:
  print("Fine = Rs.", days * 2)
elif days <= 10:
  print("Fine = Rs.", days * 3)
elif days <= 30:
  print("Fine = Rs.", days * 5)
else:
  print("Membership Cancelled.")
```

RESULT: THE PROGRAM IS COMPILED SUCCESSFULLY AND THE OUTPUT IS VERIFIED.

OUTPUT

```
Enter the number of terms: 5
0 1 1 2 3
```

PROGRAM-4

FIBONACCI SERIES WITH IN A LIMIT

AIM: Python program to print the first n terms of the Fibonacci series.

SOURCE CODE:

```
n = int(input("Enter the number of terms: "))
a = 0
b = 1
for i in range(n):
  print(a, end=" ")
  c = a + b
  a = b
  b = c
```

RESULT: THE PROGRAM IS COMPILED SUCCESSFULLY AND THE OUTPUT IS VERIFIED.

OUTPUT

```
Enter starting number: 2
Enter ending number: 31
Prime numbers are:
2
3
5
7
11
13
17
19
23
29
31
```

PROGRAM-5

PRIME NUMBERS WITHIN A RANGE

AIM: Python program to find all prime numbers within a given range.

SOURCE CODE:

```
start = int(input("Enter starting number: "))
end = int(input("Enter ending number: "))
print("Prime numbers are:")
for num in range(start, end + 1):
  if num > 1:
    for i in range(2, num):
      if num % i == 0:
        break
    else:
      print(num)
```

RESULT: THE PROGRAM IS COMPILED SUCCESSFULLY AND THE OUTPUT IS VERIFIED.

OUTPUT

```
Today's Date: 2026-07-13
Current Time: 20:21:41
```

PROGRAM-6

DISPLAY TODAY'S DATE & CURRENT TIME

AIM: Python Program to print date, time for today and now.

SOURCE CODE:

```
from datetime import datetime, time

now = datetime.now()

print("Today's Date:", now.date())
print("Current Time:", now.strftime("%H:%M:%S"))
```

RESULT: THE PROGRAM IS COMPILED SUCCESSFULLY AND THE OUTPUT IS VERIFIED.

OUTPUT

```
After append: [10, 20, 30, 40]
After insert: [10, 15, 20, 30, 40]
After remove: [10, 15, 30, 40]
Length of list: 4
After pop: [10, 15, 30]
After clear: []
```

PROGRAM-7

PYTHON LIST OPERATIONS

AIM: Python program to Create a list and perform following methods: insert(), remove(), append(), len(),pop(), clear().

SOURCE CODE:

```
numbers = [10, 20, 30]
# append
numbers.append(40)
print("After append:", numbers)

# insert
numbers.insert(1, 15)
print("After insert:", numbers)

# remove
numbers.remove(20)
print("After remove:", numbers)

# len
print("Length of list:", len(numbers))

# pop
numbers.pop()
print("After pop:", numbers)

# clear
numbers.clear()
print("After clear:", numbers)
```

RESULT: THE PROGRAM IS COMPILED SUCCESSFULLY AND THE OUTPUT IS VERIFIED.

OUTPUT

```
Enter first number: 32
Enter second number: 6
Result = 5.333333333333333

=== RESTART: C:\Users\print\AppData\Local\P
Enter first number: 13
Enter second number: d
Error: Please enter valid integers.
```

PROGRAM-8

ARITHMETIC DIVISION USING MULTIPLE EXCEPTION

AIM: Python program to include multiple exceptions.

SOURCE CODE:

```
try:
    a = int(input("Enter first number: "))
    b = int(input("Enter second number: "))

    result = a / b
    print("Result =", result)

except ZeroDivisionError:
    print("Error: Cannot divide by zero.")

except ValueError:
    print("Error: Please enter valid integers.")

except Exception as e:
    print("An unexpected error occurred:", e)
```

RESULT: THE PROGRAM IS COMPILED SUCCESSFULLY & THE OUTPUT IS VERIFIED.

OUTPUT

```
Enter numerator: 25
Enter denominator: 18
Result = 1.3888888888888888

=== RESTART: C:\Users\print\AppData\Local\Progr
Enter numerator: 5
Enter denominator: 0
Error: Division by zero is not allowed.
```

PROGRAM-9

EXCEPTION HANDLING USING DIVISION BY ZERODIVISIONERROR

AIM: Python program to demonstrate the exception handling for 0.

SOURCE CODE:

```
try:
    num1 = int(input("Enter numerator: "))
    num2 = int(input("Enter denominator: "))

    result = num1 / num2
    print("Result =", result)

except ZeroDivisionError:
    print("Error: Division by zero is not allowed.")
```

RESULT: THE PROGRAM IS COMPILED SUCCESSFULLY AND THE OUTPUT IS VERIFIED.

```
Enter a number: 5
Factorial = 120
```

PROGRAM-10

FACTORIAL OF A NUMBER USING FUNCTION

AIM: Python program to find factorial of a number using functions.

SOURCE CODE:

```
def factorial(n):
    fact = 1
    for i in range(1, n + 1):
        fact *= i
    return fact

num = int(input("Enter a number: "))

if num < 0:
    print("Factorial does not exist for negative numbers.")
else:
    print("Factorial =", factorial(num))
```

RESULT: THE PROGRAM IS COMPILED SUCCESSFULLY & THE OUTPUT IS VERIFIED.

OUTPUT

```
Menu
+ : Addition
- : substraction
* : multiplication
/ : division
0 : to exit
Enter value 1st value: 12
Enter value 2nd value: 3
Enter your choice: +
addition of 12 + 3 =  15
-----------------------
Menu
+ : Addition
- : substraction
* : multiplication
/ : division
0 : to exit
Enter value 1st value: 10
Enter value 2nd value: 5
Enter your choice: -
Subtraction of 10 - 5 =  5
```

PROGRAM-11

ARITHMETIC OPERATIONS USING FUNCTION

AIM: Python program to create to menu to perform addition, subtraction, multiplication, division.

SOURCE CODE:

```
def addition(a,b):
  return a + b

def subtraction(a,b):
  return a - b

def multiplication(a,b):
  return a * b

def division(a,b):
  try:
    return a / b
  except ZeroDivisionError as e:
    print("!!!! Error: ",e )
while(1):

  print("Menu\n+ : Addition\n- : substraction\n* : multiplication\n/ : division\n0 : to exit")
  x = int(input("Enter value 1st value: "))
  y = int(input("Enter value 2nd value: "))
  choice = input("Enter your choice: ")
  match choice:
    case '+':
      print(f'addition of {x} + {y} = ' ,addition(x,y))
    case '-':
      print(f'Subtraction of {x} - {y} = ' ,subtraction(x,y))
```

OUTPUT

```
Menu
+ : Addition
- : substraction
* : multiplication
/ : division
0 : to exit
Enter value 1st value: 5
Enter value 2nd value: 5
Enter your choice: *
Product of 5 * 5 = 25
-----------------------
Menu
+ : Addition
- : substraction
* : multiplication
/ : division
0 : to exit
Enter value 1st value: 10
Enter value 2nd value: 2
Enter your choice: /
Division of 10 / 2 = 5.0
-----------------------
```

```
  case '*':
    print(f'Product of {x} * {y} =' ,multiplication(x,y))
  case '/':
    print(f'Division of {x} / {y} =' ,division(x,y))
  case '0':
    break
  case _:
    print("Invalid Expression...")

print("----------------")
```

RESULT: THE PROGRAM IS COMPILED SUCCESSFULLY AND THE OUTPUT IS VERIFIED.

OUTPUT

```
Model:  BMW
Maximum Speed:  120 km/hrs
Mileage:  23 km/lrs
```

PROGRAM-12

VEHICLE CLASS WITH MAXIMUM SPEED & MILEAGE

AIM: Python program to create a vehicle class with maximum speed, mileages as instant attribute.

SOURCE CODE:

```
class Vehicle:
  def __init__(self,name,max_speed,mileage):
    self.name = name
    self.speed = max_speed
    self.mileage = mileage

car = Vehicle("BMW",120,23)

print("Model: ", car.name)
print("Maximum Speed: ", car.speed ,"km/hrs")
print("Mileage: ", car.mileage ,"km/lrs")
```

RESULT: THE PROGRAM IS COMPILED SUCCESSFULLY AND THE OUTPUT IS VERIFIED.

OUTPUT

```
Constructor Called...!
Student Name is  Naruto Utsumaki
Object created
Destructor Called....!
Object deleted
```

PROGRAM-13

CONSTRUCTOR & DESTRUCTOR DEMONSTRATION

AIM: Python program to demonstrate the concept of constructor & destructor.

SOURCE CODE:

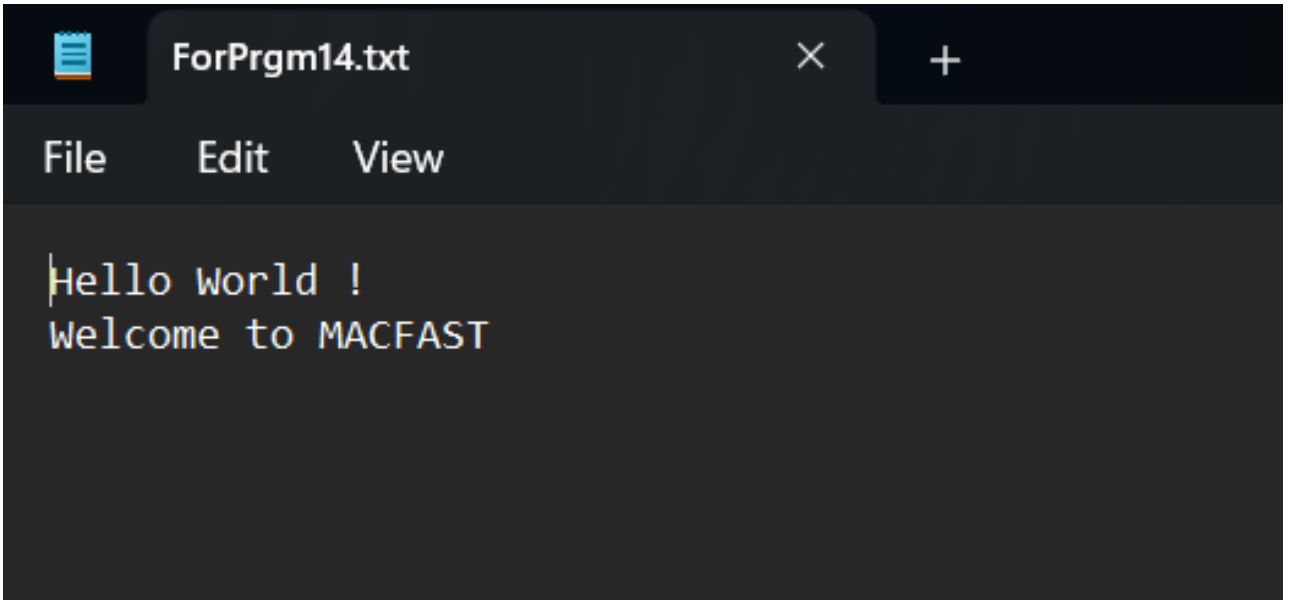
```
class Student:
  def __init__(self,name):
    self.name = name
    print("Constructor Called...!")
    print("Student Name is ",self.name)
    print("Object created")

  def __del__(self):
    print("Destructor Called. .. !")
    print("Object deleted")


s1 = Student("Naruto Utsumaki")
del s1
```

RESULT: THE PROGRAM IS COMPILED SUCCESSFULLY AND THE OUTPUT IS VERIFIED.

OUTPUT




```
Line- 1
Hello World !

No : of characters = 14
Line- 2
Welcome to MACFAST

No : of characters = 19
Longest Line: Welcome to MACFAST
 with characters = 19
```

PROGRAM-14

FUNCTIONS TO FIND THE LONGEST LINE IN A FILE

AIM: Write a program with function to find the "largest" which accepts a filename as a parameter and reports the longest line in the file.

SOURCE CODE:

```
def findLarge(f):
    long = ""
    count = 0
    for line in f:
        count += 1
        print('Line-',count)
        print(line)
        print('No : of characters =', len(line))
        if len(line) > len(long):
            long = line
    print('Longest Line:', long, 'with characters =', len(long))

f = open("ForPrgm14.txt", "r")
findLarge(f)
```

RESULT: THE PROGRAM IS COMPILED SUCCESSFULLY AND THE OUTPUT IS VERIFIED.

OUTPUT

```
Menu
+ : Addition
- : substraction
* : multiplication
/ : division
0 : to exit
Enter value 1st value: 12
Enter value 2nd value: 3
Enter your choice: +
addition of 12 + 3 =  15
----------------------
Menu
+ : Addition
- : substraction
* : multiplication
/ : division
0 : to exit
Enter value 1st value: 10
Enter value 2nd value: 5
Enter your choice: -
Subtraction of 10 - 5 =  5
```

PROGRAM-15

CREATING & IMPORTING A CALCULATOR MODULE

AIM: Python program to create and import a calculator module name 'calc.py'

SOURCE CODE:

calc.py

```
def addition(a,b):
  return a + b
def subtraction(a,b):
  return a - b
def multiplication(a,b):
  return a * b
def division(a,b):
  try:
    return a / b
  except ZeroDivisionError as e:
    print("!!!! Error: ",e )
```

main.py

```
import calc
while(1):
  print("Menu\n+ : Addition\n- : substraction\n* : multiplication\n/ : division\n0 : to exit")
  x = int(input("Enter value 1st value: "))
  y = int(input("Enter value 2nd value: "))
  choice = input("Enter your choice: ")
  match choice:
    case '+':
      print(f'addition of {x} + {y} = ' ,calc.addition(x,y))
    case '-':
      print(f'Subtraction of {x} - {y} = ' ,calc.subtraction(x,y))
```

OUTPUT

```
Menu
+ : Addition
- : substraction
* : multiplication
/ : division
0 : to exit
Enter value 1st value: 5
Enter value 2nd value: 5
Enter your choice: *
Product of 5 * 5 = 25
-----------------------
Menu
+ : Addition
- : substraction
* : multiplication
/ : division
0 : to exit
Enter value 1st value: 10
Enter value 2nd value: 2
Enter your choice: /
Division of 10 / 2 = 5.0
-----------------------
```

```
  case '*':
    print(f'Product of {x} * {y} =' ,calc.multiplication(x,y))
  case '/':
    print(f'Division of {x} / {y} =' ,calc.division(x,y))
  case '0':
    break
  case _:
    print("Invalid Expression...")

print("----------------")
```

RESULT: THE PROGRAM IS COMPILED SUCCESSFULLY AND THE OUTPUT IS VERIFIED.

OUTPUT

Output - 1

```
Opened database
Table created
```

Output – 2

```
Opened database
enter rollno(enter 0 for exit):1
Enter name: Jiyo
Enter mark: 50
insert into student values(1,'Jiyo',50)


Records inserted
enter rollno(enter 0 for exit):2
Enter name: Jomy
Enter mark: 60
insert into student values(2,'Jomy',60)


Records inserted
enter rollno(enter 0 for exit):3
Enter name: Rahul
Enter mark: 86
insert into student values(3,'Rahul',86)


Records inserted
enter rollno(enter 0 for exit):0
```

PROGRAM – 16

SQLITE OPERATIONS

AIM: Develop programs to perform operations on SQLITE3

SOURCE CODE:

1) Create a Sqlite

```
import sqlite3
conn=sqlite3.connect('macfast.db')
print("Opened database")
sql="create table student(rollno int(3),name varchar(30),mark int(3))"
conn.execute(sql)
print("Table created")
conn.close()
```

2) Insert Data into Sqlite

```
import sqlite3
conn=sqlite3.connect('macfast.db')
print("Opened database")
while 1:
   rollno=input("enter rollno(enter 0 for exit):")
   if int(rollno)==0:
      break
   name=input("Enter name:")
   mark=input("Enter mark")
   sql="insert into student values(%s,'%s',%s)\n"%(rollno,name,mark)
   print(sql)
   conn.execute(sql)
   conn.commit()
   print("\nRecords inserted")
   conn.close()
```

OUTPUT

Output – 3

```
opened database
displaying data

............
Rollno|Name|Marks

............
     1|Jiyo K Thomas  | 85
     2|Jomy Joy       | 67
     3|Rahul Thomas   | 72
     4|Akhiljith S    |100
     5|Yedhu R        | 79

...........
```

Output – 4

```
opened database
Deleting data

...........
Rollno|Name|Marks

...........
(1, 'Jiyo K Thomas', 85)
(2, 'Jomy Joy', 67)
(4, 'Akhiljith S', 100)
(5, 'Yedhu R', 79)

.........
```

3) Display Data in Sqlite

```
import sqlite3
conn=sqlite3.connect('macfast.db')
print('opened database')
print('displaying     data')
print("\n.....")
result=conn.execute("select * from student")
print("Rollno|Name|Marks")
print("\n.....")
for i in result:
    print("%6s|%-15s|%3s"%(i[0],i[1],i[2]))
print("\n.....")
conn.close()
```

4) Delete Data in Sqlite

```
import sqlite3
conn=sqlite3.connect('macfast.db')
print("opened database")
print("Deleting      data")
print("\n.....")
result=conn.execute("'DELETE from student where rollno=3;'")
print("Rollno|Name|Marks")
print("\n.....")
data=conn.execute("select * from student")
for i in data:
    print(i)
print("\n.....")
conn.commit()
conn.close()
```

RESULT: THE PROGRAM IS COMPILED SUCCESSFULLY AND THE OUTPUT IS OBTAINED

OUTPUT

Output-1

```
0    2
1    4
2    5
3    6
4    9
dtype: int64
```

Output-2

```
1    2
3    4
5    5
7    6
9    9
dtype: int64
```

Output-3

```
DatetimeIndex(['2026-06-15', '2026-06-16', '2026-06-17', '2026-06-18',
               '2026-06-19', '2026-06-20', '2026-06-21', '2026-06-22',
               '2026-06-23', '2026-06-24', '2026-06-25', '2026-06-26',
               '2026-06-27', '2026-06-28', '2026-06-29', '2026-06-30',
               '2026-07-01', '2026-07-02', '2026-07-03', '2026-07-04',
               '2026-07-05', '2026-07-06', '2026-07-07', '2026-07-08',
               '2026-07-09', '2026-07-10', '2026-07-11', '2026-07-12',
               '2026-07-13', '2026-07-14', '2026-07-15', '2026-07-16',
               '2026-07-17', '2026-07-18', '2026-07-19', '2026-07-20',
               '2026-07-21', '2026-07-22', '2026-07-23', '2026-07-24',
               '2026-07-25', '2026-07-26', '2026-07-27'],
              dtype='datetime64[ns]', freq='D')
```

PROGRAM – 17

DATA PREPROCESSING TOOLKIT WITH PANDAS & NUMPY

➔ PANDAS

AIM: Develop programs for data preprocessing with pandas

1) List To Series Conversion.

SOURCE CODE:

```
import pandas as pd
a=[2,4,5,6,9]
s=pd.Series(a)
print(s)
```

2) List to Series conversion with Custom Indexing.

SOURCE CODE:

```
import pandas as pd
a=[2,4,5,6,9]
s=pd.Series(a,index=[1,3,5,7,9])
print(s)
```

3) Data Series Generation

SOURCE CODE:

```
import pandas as pd
s=pd.date_range(start='15-06-2026',end='27-07-2026')
print(s)
```

OUTPUT

Output-4

```
0     2
1     4
2     6
3     8
4    10
dtype: int64

0    1.0
1    2.0
2    3.0
3    4.0
4    5.0
dtype: float64
```

Output-5

	name	age	occ
0	Vinay	22	Engineer
1	Kushal	25	Doctor
2	Aman	24	Accountant

Output-6

	Id	Name	Age
0	2	Vishal	22
1	1	Kushal	25
2	1	Aman	24

4) Implementing a function on each and every element of a series.

SOURCE CODE:

```
import pandas as pd
s=pd.Series([2,4,6,8,10])
print(s)
print()
m_s=s.apply(lambda x:x/2)
print(m_s)
```

5) Dictionary to Dataframe Conversion.

SOURCE CODE:

```
import pandas as pd
d={'name':['Vinay','Kushal','Aman' ],'age':[22,25,24],'occ':['Engineer','Doctor','Accountant']}
df=pd.DataFrame(d)
print(df)
```

6) 2D List-to-DataFrame Conversion

SOURCE CODE:

```
import pandas as pd
l=[[2,'Vishal',22],[1,'Kushal',25],[1,'Aman',24]]
df=pd.DataFrame(l,columns=['Id','Name','Age'])
print(df)
```

RESULT: THE PROGRAM IS COMPILED SUCCESSFULLY AND THE OUTPUT IS VERIFIED.

OUTPUT

Output-1

```
Original Array:
[1 2 3 4]
Test if none of the elements of the said array is zero:
True
Original Array:
[0 1 2 3]
Test if none of the elements of the said array is zero:
False
```

Output-2

```
Original Array:
[1 0 0 0]
Test whether any of the elements of a given array is non zero:
True
Original Array:
[0 0 0 0]
Test any of the elements of a given array is non zero:
False
```

➔ NUMPY

AIM: Develop programs for numeric analysis using numpy.

1) NumPy program to test whether none of the elements of a given array is zero.

SOURCE CODE:

```
import numpy as np
x = np.array([1,2,3,4])
print("Original Array:")
print(x)
print("Test if none of the elements of the said array is zero:")
print(np.all(x))
x = np.array([0,1,2,3])
print("Original Array:")
print(x)
print("Test if none of the elements of the said array is zero:")
print(np.all(x))
```

2) NumPy program to test whether any of the elements of a given array is non-zero.

SOURCE CODE:

```
import numpy as np
x = np.array([1,0,0,0])
print("Original Array:")
print(x)
print("Test whether any of the elements of a given array is non zero:")
print(np.any(x))
x = np.array([0,0,0,0])
print("Original Array:")
print(x)
print("Test any of the elements of a given array is non zero:")
print(np.any(x))
```

OUTPUT

Output-3

```
Original Numbers:
[3 5]
[2 5]
Comparison-greater
[ True False]
Comparison-greater_equal
[ True  True]
Comparison-less
[False False]
Comparison-greater
[False  True]
```

Output-4

```
Original Array:
[  1   7  13 105]
Size of the memory occupied by the said array:
32 bytes
```

3) NumPy program to create an element-wise comparison (greater, greater_equal, less and less_equal) of two given arrays.

SOURCE CODE:

```
import numpy as np
x = np.array([3,5])
y = np.array([2,5])
print("Original Numbers:")
print(x)
print(y)
print("Comparison-greater")
print(np.greater(x,y))
print("Comparison-greater_equal")
print(np.greater_equal(x,y))
print("Comparison-less")
print(np.less(x,y))
print("Comparison-greater")
print(np.less_equal(x,y))
```

4) NumPy program to create an array with the values 1,7,13,105 and determine the size of the memory occupied by the array.

SOURCE CODE:

```
import numpy as np
X = np.array([1,7,13,105])
print("Original Array:")
print(X)
print("Size of the memory occupied by the said array:")
print("%d bytes" %(X.size * X.itemsize))
```

OUTPUT

Output-5

```
Original List: [12.23, 13.32, 100, 36.32]
One-dimensional NumPy array: [ 12.23  13.32 100.    36.32]
```

Output-6

```
[[ 2  3  4]
 [ 5  6  7]
 [ 8  9 10]]
```

Output-7

```
[12 13 14 15 16 17 18 19 20 21 22 23 24 25 26 27 28 29 30 31 32 33 34 35
 36 37]
```

Output-8

```
[12 13 14 15 16 17 18 19 20 21 22 23 24 25 26 27 28 29 30 31 32 33 34 35
 36 37]
Reverse Array:
[37 36 35 34 33 32 31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14
 13 12]
```

5) NumPy program to convert a list of numeric value (12.23,13.32, 100,36.32) into a one-dimensional NumPy array.

SOURCE CODE:

```
import numpy as np
I = [12.23,13.32,100,36.32]
print("Original List:",I)
a=np.array(I)
print("One-dimensional NumPy array:",a)
```

6) NumPy program to create a 3x3 matrix with values ranging from 2 to 10.

SOURCE CODE:

```
import numpy as np
x= np.arange(2,11).reshape(3,3)
print(x)
```

7) NumPy program to create an array with values ranging from 12 to 38.

SOURCE CODE:

```
import numpy as np
x= np.arange(12,38)
print(x)
```

8) NumPy program to reverse an array (first element becomes last).

SOURCE CODE:

```
import numpy as np
x= np.arange(12,38)
print(x)
print("Reverse Array:")
x=x[::-1]
print(x)
```

OUTPUT

Output-9

```
Original array:
[[1. 1. 1. 1. 1.]
 [1. 1. 1. 1. 1.]
 [1. 1. 1. 1. 1.]
 [1. 1. 1. 1. 1.]
 [1. 1. 1. 1. 1.]]
1 on the border and 0 inside in the array
[[1. 1. 1. 1. 1.]
 [1. 0. 0. 0. 1.]
 [1. 0. 0. 0. 1.]
 [1. 0. 0. 0. 1.]
 [1. 1. 1. 1. 1.]]
```

Output-10

```
Original array:
[10, 20, 30]
After append values to the end of the array:
[10 20 30 40 50 60 70 80 90]
```

9) NumPy program to create a 2d array with 1 on the border and 0 inside as shown in the picture below.

SOURCE CODE:

```
import numpy as np
x= np.ones((5,5))
print("Original array:")
print(x)
print("1 on the border and 0 inside in the array")
x[1:-1,1:-1]=0
print(x)
```

10) NumPy program to append values to the end of an array.

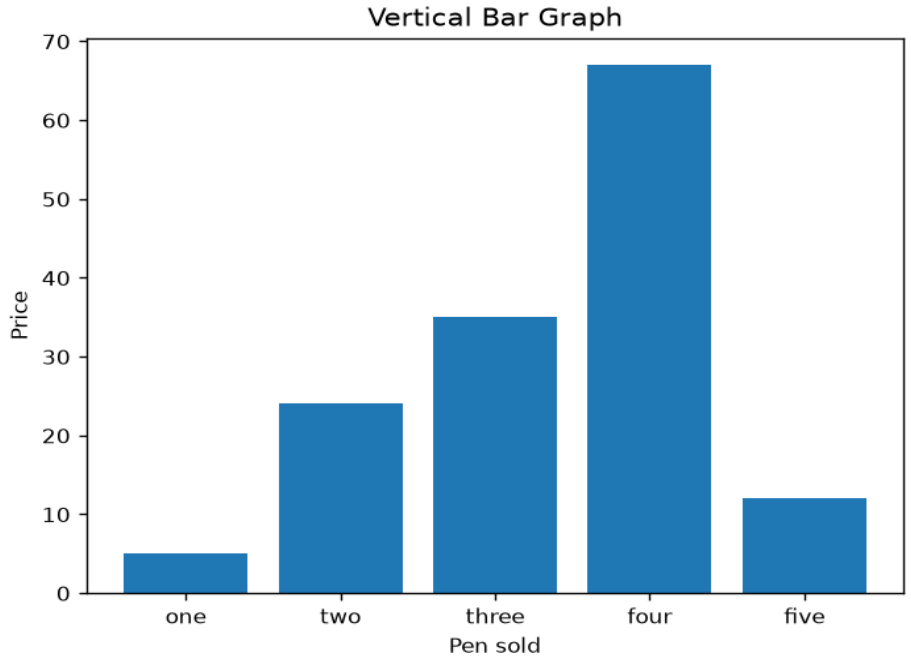
SOURCE CODE:

```
import numpy as np
x= [10,20,30]
print("Original array:")
print(x)
x=np.append(x,[[40,50,60],[70,80,90]])
print("After append values to the end of the array:")
print(x)
```

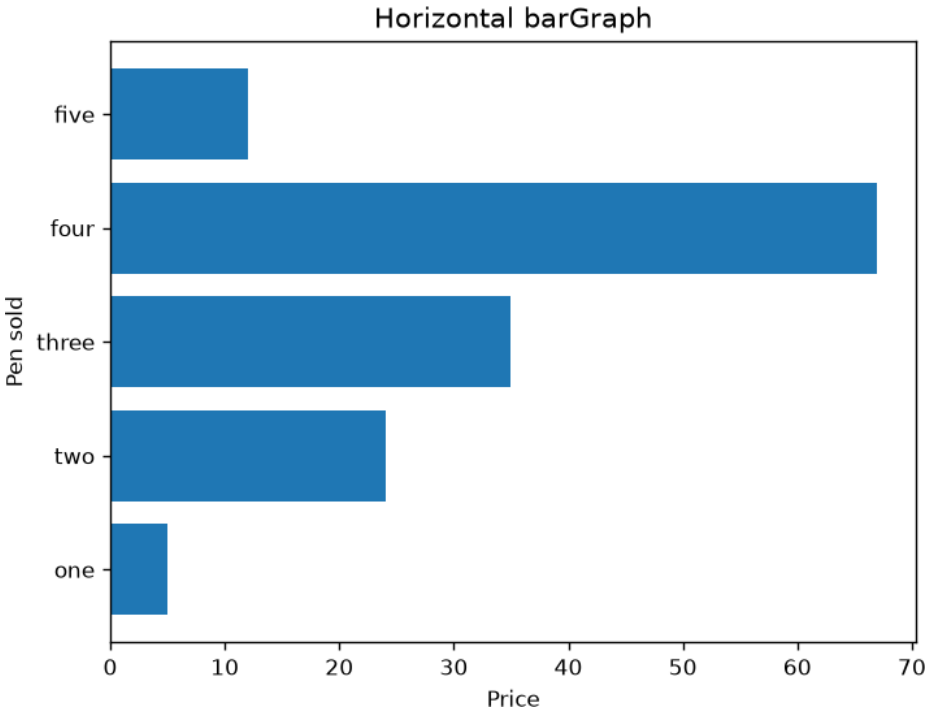
RESULT: THE PROGRAM IS COMPILED SUCCESSFULLY AND THE OUTPUT IS VERIFIED.

OUTPUT

Output-1



Output-2



PROGRAM – 18

DATA PREPROCESSING TOOLKIT WITH MATPLOTLIB

AIM: Develop programs for implementing plots with matplotlib.

1) Creating a vertical bar chart

SOURCE CODE:

```
import matplotlib.pyplot as plt
x= ['one','two','three','four','five']
y = [5,24,35,67,12]
plt.bar(x,y)
plt.xlabel("Pen sold")
plt.ylabel("Price")
plt.title("Vertical Bar Graph")
plt.show()
```

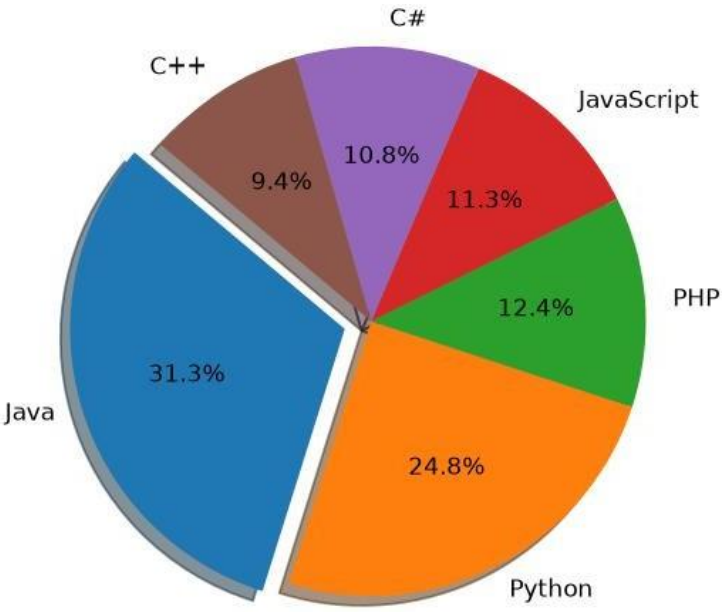
2) Creating a horizontal bar chart

SOURCE CODE:

```
import matplotlib.pyplot as plt
y= ['one','two','three','four','five']
x = [5,24,35,67,12]
plt.barh(y,x)
plt.ylabel("Pen sold")
plt.xlabel("Price")
plt.title("Horizontal barGraph")
plt.show()
```

OUTPUT

Output-3



3) Write a Python program to create a pie chart of the popularity of programming languages.

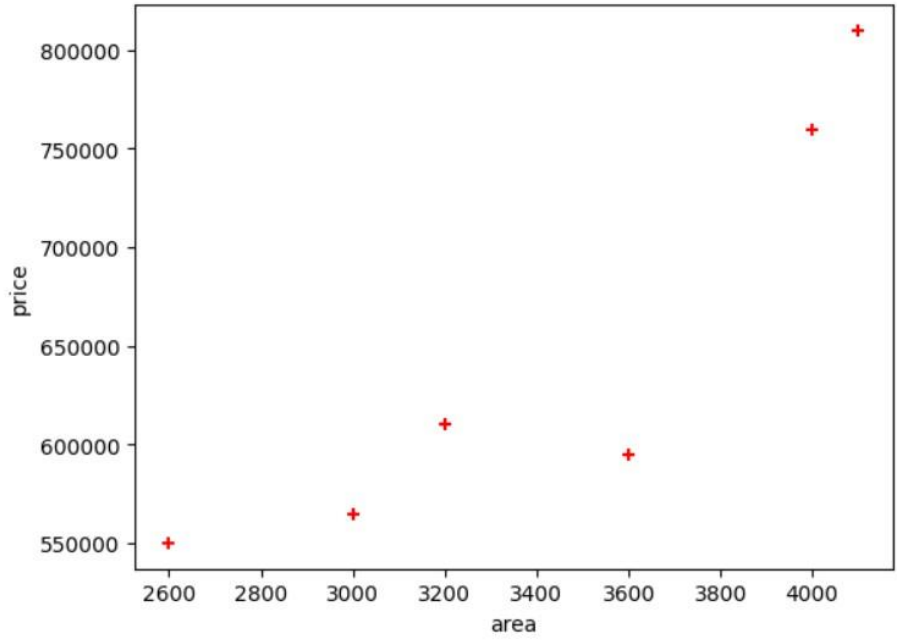
SOURCE CODE:

```
import matplotlib.pyplot as plt
l='Java','Python','PHP','JavaScript','C#','C++'
p=[22.2,17.6,8.8,8,7.7,6.7]
c=["#1f77b4","#ff7f0e","#2ca02c","#d62728","#9467bd","#8c564b"]
e=(0.1,0,0,0,0,0)
plt.pie(p,explode=e,labels=l,colors=c,autopct='%1.1f%%',shadow=True,startangle=140)
plt.axis('Equal')
plt.show()
```

RESULT: THE PROGRAM IS COMPILED SUCCESSFULLY AND THE OUTPUT IS VERIFIED.

OUTPUT

```
   area  bedroom  age   price
0  2600        3   20  550000
1  3000        4   15  565000
2  3200        4   18  610000
3  3600        3   30  595000
4  4000        5    8  760000
5  4100        6    8  810000
```




```
   area  bedroom  age
0  2600        3   20
1  3000        4   15
2  3200        4   18
3  3600        3   30
4  4000        5    8
5  4100        6    8
0    550000
1    565000
2    610000
3    595000
4    760000
5    810000
Name: price, dtype: int64
LinearRegression()
[498408.25158031]
```

PROGRAM-19

SIMPLE LINEAR REGRESSION

AIM: Develop programs based on linear regression.

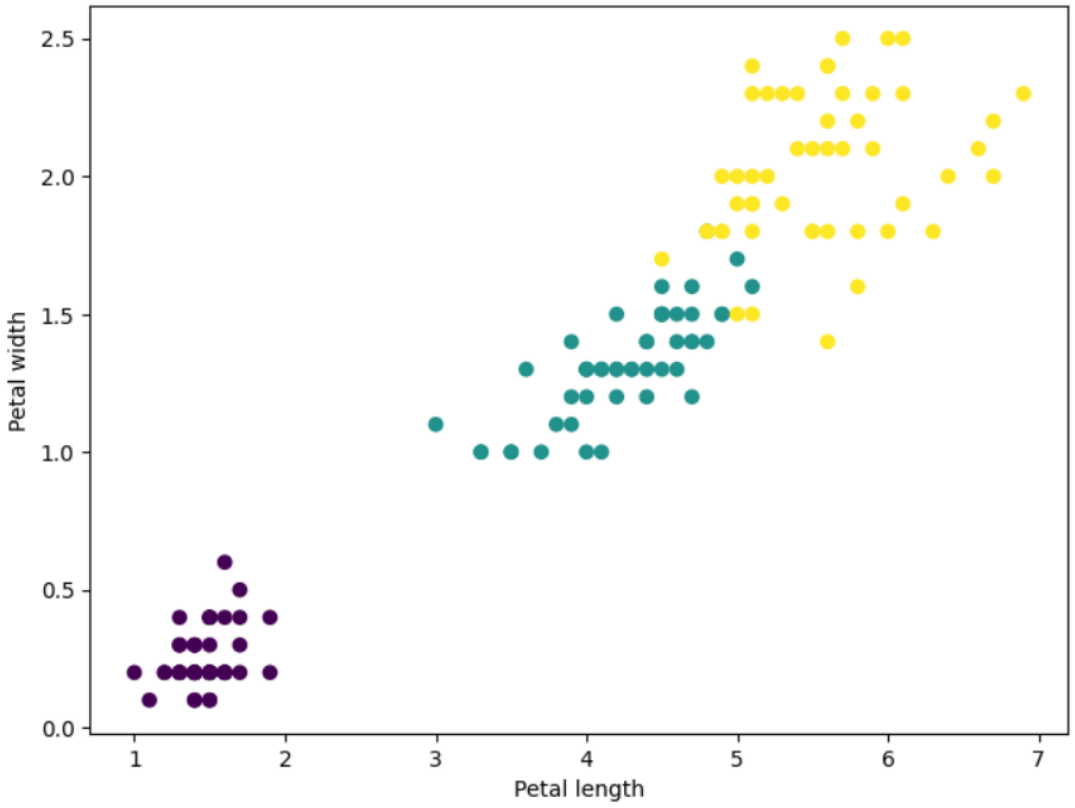
SOURCE CODE:

```
import pandas as pd
import numpy as np
from sklearn import linear_model
import matplotlib.pyplot as plt
df  = {'area':[2600,3000,3200,3600,4000,4100],
      'bedroom':[3,4,4,3,5,6],
      'age':[20,15,18,30,8,8],
      'price':[550000,565000,610000,595000,760000,810000]
      }
df = pd.DataFrame(df)
print(df)
plt.xlabel('area')
plt.ylabel('price')
plt.scatter(df.area,df.price,color='red',marker='+')
plt.show()
new_df =  df.drop('price',axis='columns')
print(new_df)
price = df.price
print(price)
reg= linear_model.LinearRegression()
print(reg.fit(df.drop('price',axis='columns').values,df.price.values))
print(reg.predict([[3000,3,40]]))
```

RESULT: THE PROGRAM IS COMPILED SUCCESSFULLY AND THE OUTPUT IS VERIFIED.

OUTPUT

```
[[1.4 0.2]
 [1.3 0.2]
 [1.5 0.2]
 [1.4 0.2]]
[0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0
 0 0 0 0 0 0 0 0 0 0 0 0 0 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1
 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 2 2 2 2 2 2 2 2 2 2 2 2 2
 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2
 2 2]
```



PROGRAM-20

CLASSIFICATION BASED LOGISTIC REGRESSION

AIM: Implementation of at least one classification algorithm using scikit-learn.

SOURCE CODE:

```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
from sklearn import datasets
from sklearn.model_selection import train_test_split
from sklearn.preprocessing import StandardScaler
from matplotlib.colors import ListedColormap
from sklearn.metrics import accuracy_score
from sklearn.linear_model import LogisticRegression

iris = datasets.load_iris()

X = iris.data[:,[2,3]]
Y = iris.target
print(X[1:5,:])
print(Y)
plt.figure(2, figsize=(8, 6))
plt.clf()

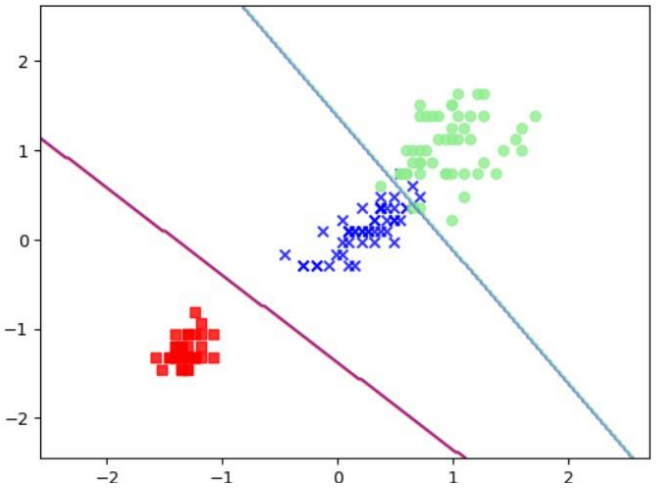
# Plot the training points
plt.scatter(X[:, 0], X[:, 1], c=Y)
plt.xlabel('Petal length')
plt.ylabel('Petal width')
plt.show()
X_train, X_test, Y_train, Y_test = train_test_split(X,Y,test_size=0.3,random_state=0)
length_Train = len(X_train)
length_Test = len(X_test)
```

OUTPUT

```
There are  105 samples in the trainig set and 45 samples in the test set
--------------------------------------------------------------------------------------------------

X_train without standardising features
---------------------------------------
[[5.5 1.8]
 [5.7 2.5]
 [5.  1.5]
 [5.8 1.8]]

X_train standardising features
---------------------------------------
[[0.93066067 0.7372463 ]
 [1.04202177 1.63887031]
 [0.6522579  0.35083601]
 [1.09770233 0.7372463 ]]
LogisticRegression(C=1000.0, random_state=5)
Accuracy: 0.98
/tmp/ipykernel_2445/153140556.py:63: UserWarning: *c* argument looks like a single numeric RGB or RGBA sequence, w
  plt.scatter(x=X[y == cl, 0], y=X[y == cl, 1],
```



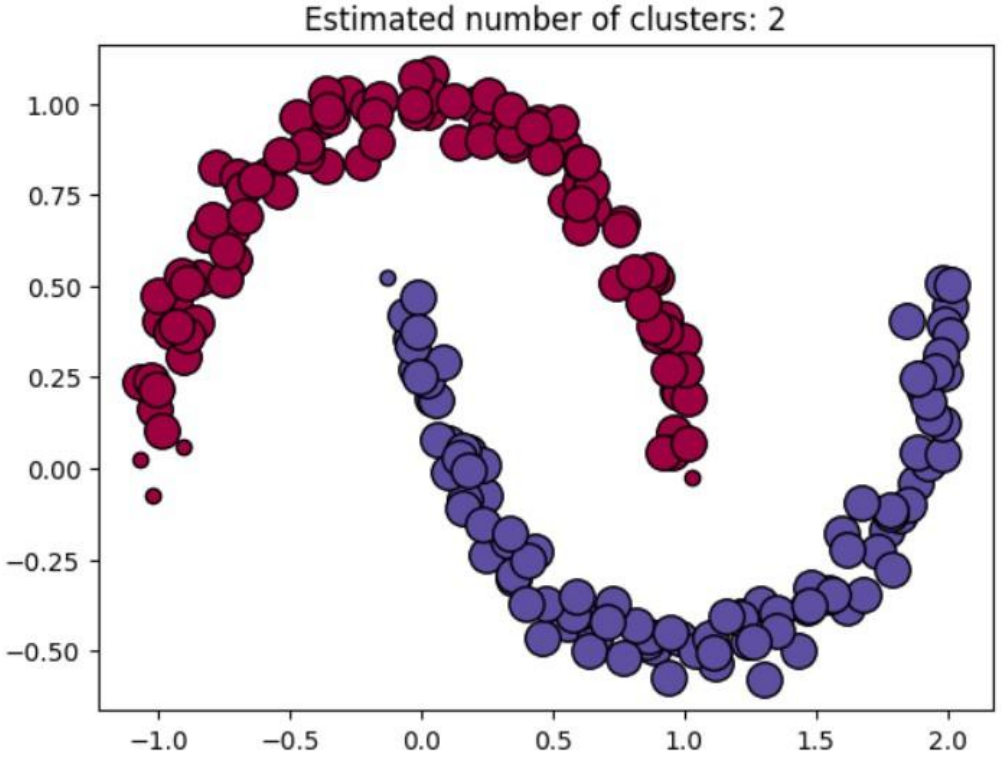
```
print("There are ",length_Train,"samples in the trainig set and",length_Test,"samples in the test set")
print("----------------------------------------------------------------")
print("")
sc = StandardScaler()
sc.fit(X_train)
X_train_standard = sc.transform(X_train)
X_test_standard = sc.transform(X_test)
print("X_train without standardising features")
print("-------------------------")
print(X_train[1:5,:])
print("")
print("X_train standardising features")
print("-------------------------")
print(X_train_standard[1:5,:])
def plot_decision_regions(X, y, classifier, test_idx=None, resolution=0.02):
    markers = ('s', 'x', 'o', '^', 'v')
    colors = ('red', 'blue', 'lightgreen', 'gray', 'cyan')
    color_Map = ListedColormap(colors[:len(np.unique(y))])
    x1_min = X[:, 0].min() - 1
    x1_max = X[:, 0].max() + 1
    x2_min = X[:, 1].min() - 1
    x2_max = X[:, 1].max() + 1
    xx1, xx2 = np.meshgrid(np.arange(x1_min, x1_max, resolution), np.arange(x2_min, x2_max, resolution))
    Z = classifier.predict(np.array([xx1.ravel(), xx2.ravel()]).T)
    Z = Z.reshape(xx1.shape)
    plt.contour(xx1, xx2, Z, alpha=0.4, cmap=color_Map)
    plt.xlim(xx1.min(), xx1.max())
    plt.ylim(xx2.min(), xx2.max())
    # Plot samples
    X_test, Y_test = X[test_idx, :], y[test_idx]
    for idx, cl in enumerate(np.unique(y)):
        plt.scatter(x=X[y == cl, 0], y=X[y == cl, 1],  alpha=0.8, c=color_Map(idx), marker=markers[idx],
                    label=cl)
```

```
X_combined_standard = np.vstack((X_train_standard,X_test_standard))
Y_combined = np.hstack((Y_train, Y_test))
lr = LogisticRegression(C = 1000.0, random_state = 5 )
print(lr.fit(X_train_standard, Y_train))
Y_pred_Logit = lr.predict(X_test_standard)
print("Accuracy: %.2f" % accuracy_score(Y_test,Y_pred_Logit))
plot_decision_regions(X = X_combined_standard , y = Y_combined, classifier = lr ,
                                        test_idx = range(105,150))
plt.show()
```

RESULT: THE PROGRAM IS COMPILED SUCCESSFULLY AND THE OUTPUT IS VERIFIED.

OUTPUT

```
[0 1 1 0 1 1 0 1 0 1 0 1 1 1 0 0 0 1 0 0 1 1 0 1 0 1 1 1 1 0 0 0 1 1 0 1 1
 0 0 1 1 0 0 1 1 0 0 0 1 1 0 1 1 0 1 0 0 1 0 0 1 0 1 0 1 0 0 1 0 0 1 0 1 1
 1 0 1 0 0 1 1 0 1 1 1 0 0 0 1 1 0 0 1 0 1 1 1 1 0 1 1 1 0 0 0 1 0 0 1 0 0
 0 0 0 0 1 0 1 1 0 0 0 1 0 1 0 0 1 1 1 0 0 0 1 1 1 1 0 1 0 1 1 0 0 0 0 1 1
 0 1 1 1 0 0 1 0 1 1 0 0 1 1 0 1 1 1 0 1 1 1 0 0 0 0 1 1 1 0 0 0 1 0 1 1 1
 0 0 1 0 0 0 0 0 0 1 0 1 1 0 1]
```



PROGRAM – 21

DBSCAN USING CLUSTERING

AIM: Implementation of at least one clustering algorithm using scikit-learn.

SOURCE CODE:

```
import matplotlib.pyplot as plt
import numpy as np
from sklearn.cluster import DBSCAN
from sklearn import metrics
from sklearn.preprocessing import StandardScaler
from sklearn import datasets
from sklearn.cluster import DBSCAN
from sklearn.datasets import make_moons

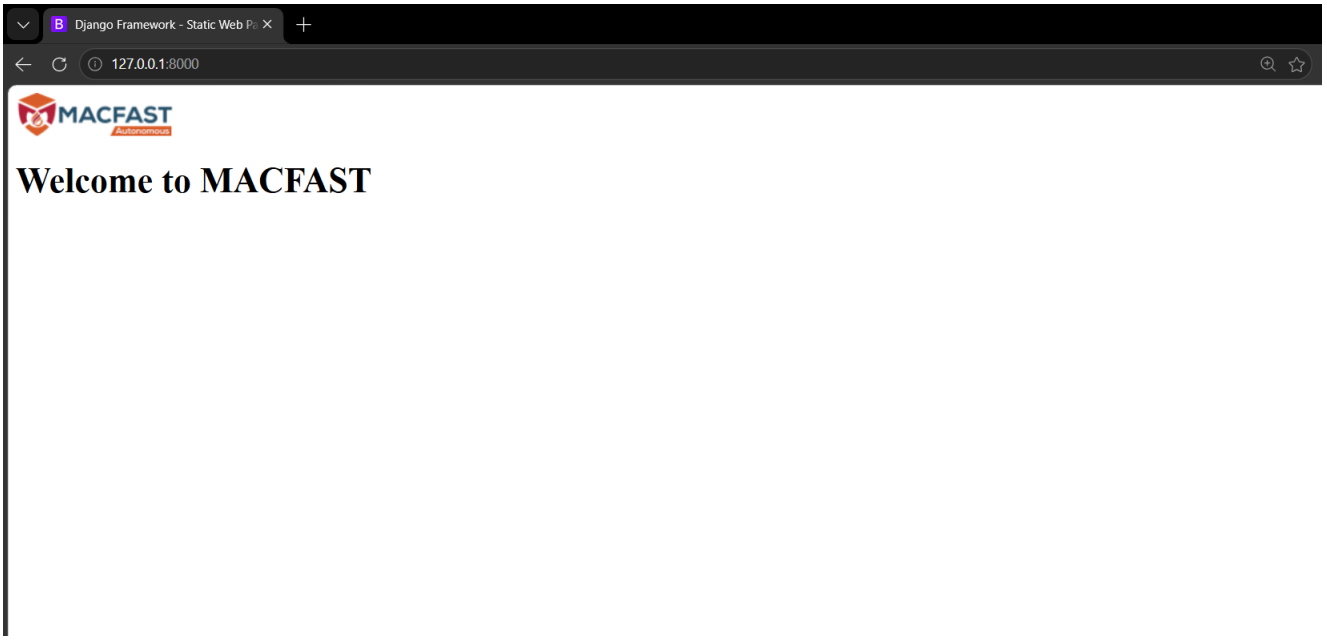
X, y = make_moons(200, noise=.05, random_state=0)
db = DBSCAN(eps=0.3, min_samples=10).fit(X)
core_samples_mask = np.zeros_like(db.labels_, dtype=bool)
core_samples_mask[db.core_sample_indices_] = True
labels = db.labels_
n_clusters_ = len(set(labels)) - (1 if -1 in labels else 0)
print(labels)

n_noise_ = list(labels).count(-1)
unique_labels = set(labels)
colors = [plt.cm.Spectral(each) for each in np.linspace(0, 1, len(unique_labels))]
for k, col in zip(unique_labels, colors):
    if k == -1:
        col = [0, 0, 0, 1]
    class_member_mask = (labels == k)
    xy = X[class_member_mask & core_samples_mask]
    plt.plot(xy[:, 0], xy[:, 1], 'o', markerfacecolor=tuple(col),markeredgecolor='k', markersize=14)
    xy = X[class_member_mask & ~core_samples_mask]
```

```
plt.plot(xy[:, 0], xy[:, 1], 'o', markerfacecolor=tuple(col),markeredgecolor='k', markersize=6)
plt.title('Estimated number of clusters: %d' % n_clusters_)
plt.show()
```

RESULT: THE PROGRAM IS COMPILED SUCCESSFULLY AND THE OUTPUT IS VERIFIED.

OUTPUT


Django Framework - Static Web
127.0.0.1:8000
MACFAST
Autonomous
Welcome to MACFAST

PROGRAM – 22

DJANGO FRAMEWORK

AIM: Develop web application with static web pages using Django framework

- **Create a Directory for the project**

```
C:\Users\print\Downloads>mkdir django-project

C:\Users\print\Downloads>cd django-project

C:\Users\print\Downloads\django-project>python -m venv env
```

- **Create the virtual environment**

```
C:\Users\print\Downloads\django-project>python -m venv env
```

- **Install Django**

```
C:\Users\print\Downloads\django-project>pip install django
Requirement already satisfied: django in C:\Users\print\AppData\Local\Programs\Python\Python312\Lib\site-packages (5.2.1)
Requirement already satisfied: asgiref>=3.8.1 in C:\Users\print\AppData\Local\Programs\Python\Python312\Lib\site-packages (from django) (3.8.1)
Requirement already satisfied: sqlparse>=0.3.1 in C:\Users\print\AppData\Local\Programs\Python\Python312\Lib\site-packages (from django) (0.5.3)
Requirement already satisfied: tzdata in C:\Users\print\AppData\Local\Programs\Python\Python312\Lib\site-packages (from django) (2024.2)

[notice] A new release of pip is available: 26.0.1 -> 26.2.1
[notice] To update, run: python.exe -m pip install --upgrade pip
```

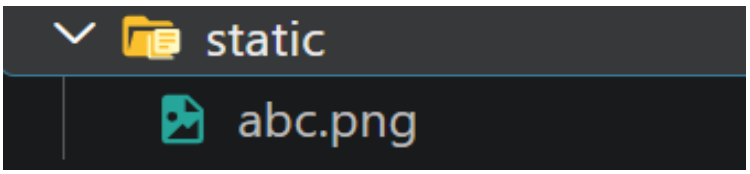
- **Create the project**

```
C:\Users\print\Downloads\django-project>django-admin startproject djgoproject .
```

- **Create an App**

```
C:\Users\print\Downloads\django-project>django-admin startapp djgoapp
```

- **Open the project in any editor and create a Directory named “static” to store the plugins and images that need to display in the website & Also insert an image in “static”.**



- **Open another Directory named “templates” & create an html page.**



- **Declare the app 'djgoapp' in INSTALLED_APPS = [ ]**

```
# Application definition
INSTALLED_APPS = [
    'django.contrib.admin',
    'django.contrib.auth',
    'django.contrib.contenttypes',
    'django.contrib.sessions',
    'django.contrib.messages',
    'django.contrib.staticfiles',
    'djgoapp'    "djgoapp": Unknown word.
]
```

- **Import os in setings.py**

```
import os
from pathlib import Path
# Build paths inside the project like this: BASE_DIR / 'subdir'.
BASE_DIR = Path(__file__).resolve().parent.parent
```

- **Add the newly created TEMPLATE_DIRS in the settings.py**

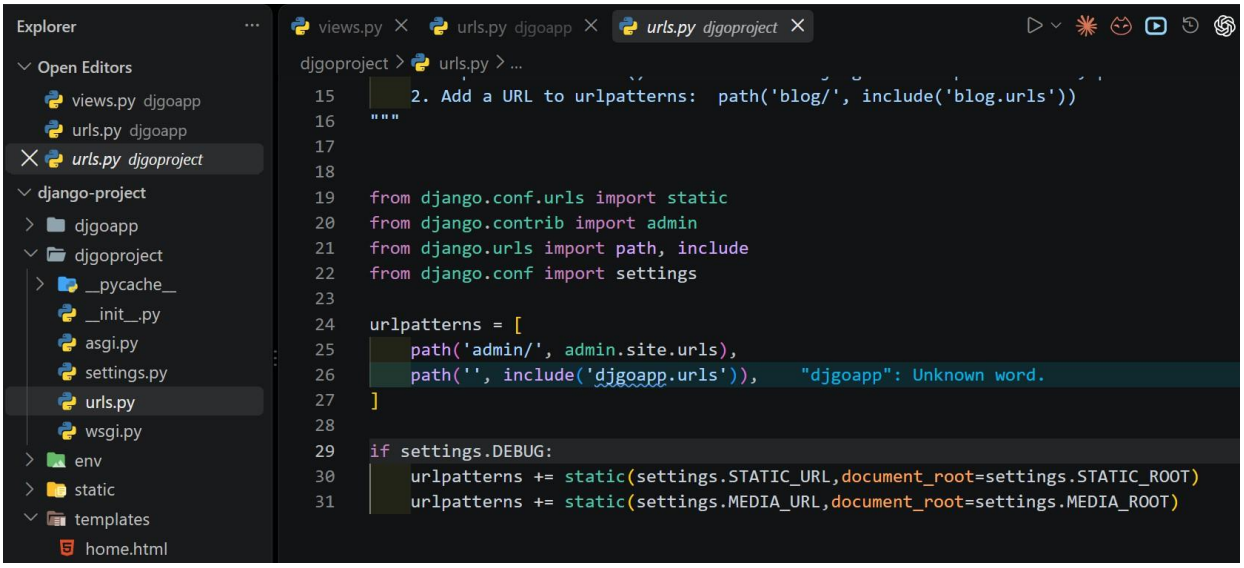
```
TEMPLATES = [
    {
        'BACKEND': 'django.template.backends.django.DjangoTemplates',
        'DIRS': [os.path.join(BASE_DIR,'templates')],
        'APP_DIRS': True,
        'OPTIONS': {
            'context_processors': [
                'django.template.context_processors.request',
                'django.contrib.auth.context_processors.auth',
                'django.contrib.messages.context_processors.messages',
            ],
        },
    },
]
```

- **Give path to STATIC_URL, STATICFILES_DIRS, MEDIA_URL, MEDIA_ROOT in settings.py**

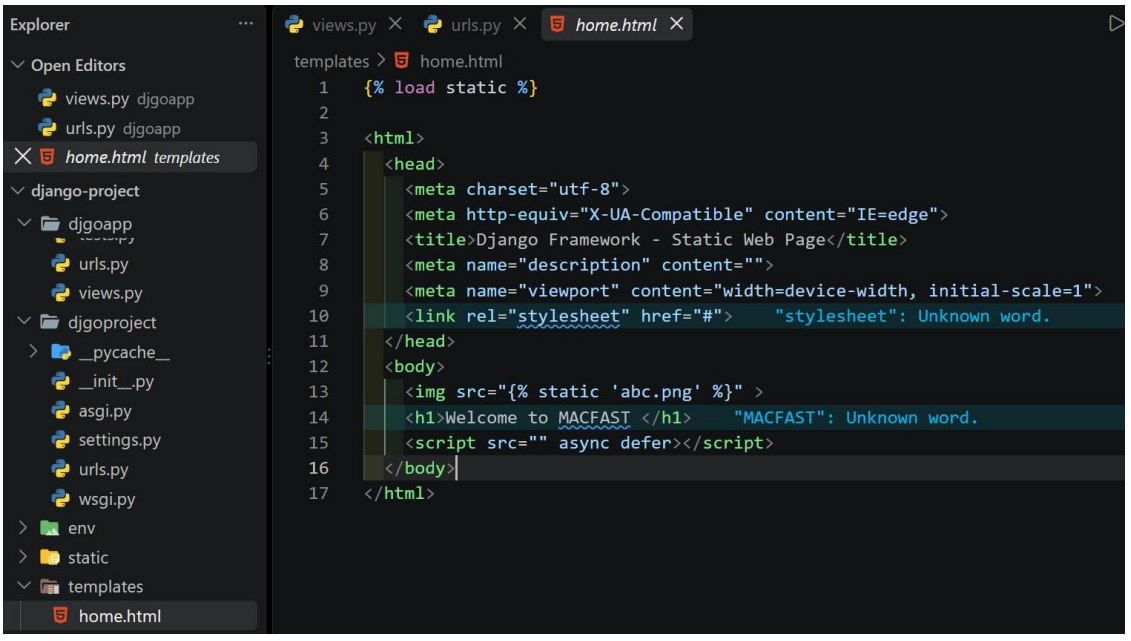
```
# Static files (CSS, JavaScript, Images)
# https://docs.djangoproject.com/en/5.2/howto/static-files/

STATIC_URL = '/static/'
STATICFILES_DIRS = [ BASE_DIR / "static" ]
MEDIA_URL = '/media/'
MEDIA_ROOT = [ BASE_DIR / "media" ]
```

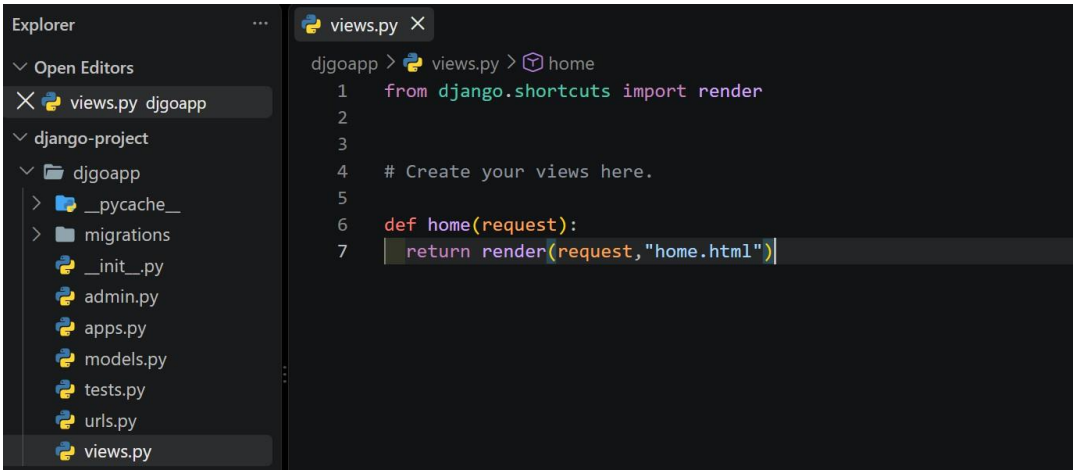
- **Give the url of static and media in the urls.py inside project**



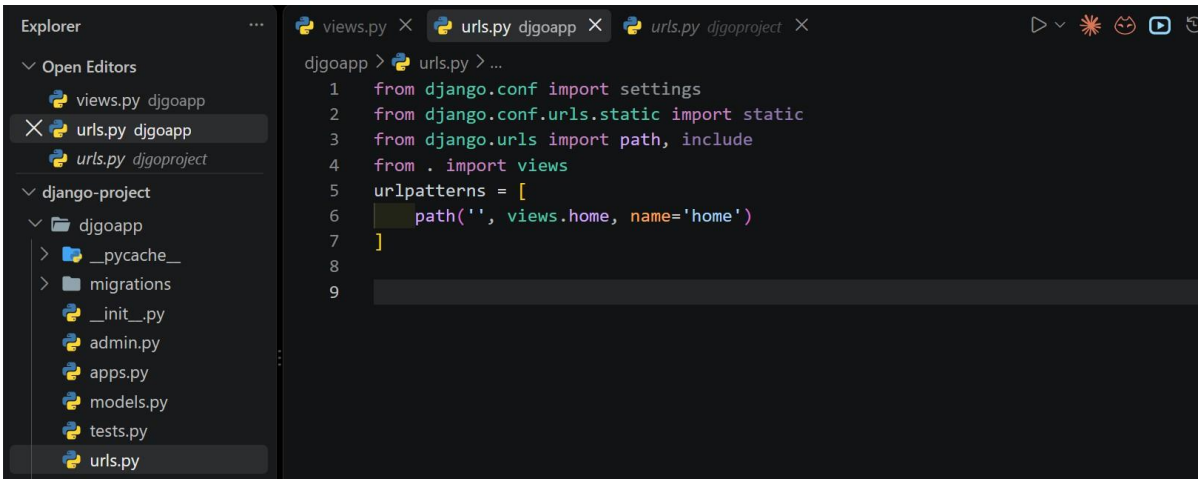
- **Load the static files, and insert the image & text inside the html file.**



➢ **Create a function inside the views.py in the app's component.**



➢ **Add urls.py in app's component.**



➢ **Run the project**

```
C:\Users\print\Downloads\django-project>python manage.py runserver
Watching for file changes with StatReloader
Performing system checks...

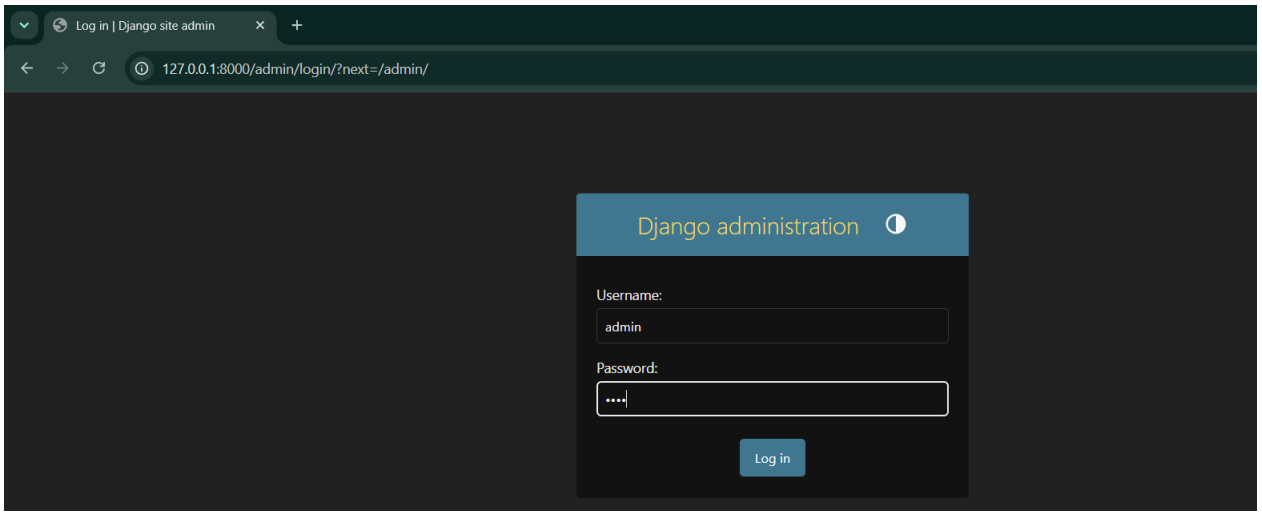
System check identified no issues (0 silenced).

You have 18 unapplied migration(s). Your project may not work properly until you apply the migrations for app(s): admin, auth, contenttypes, sessions.
Run 'python manage.py migrate' to apply them.
October 01, 2026 - 16:36:35
Django version 5.2.1, using settings 'djgoproject.settings'
Starting development server at http://127.0.0.1:8000/
Quit the server with CTRL-BREAK.
```

OUTPUT



Welcome to MACFAST



PROGRAM – 23

DJANGO FRAMEWORK

AIM: Develop web application with dynamic web pages using Django framework

- **Create a folder**
- **Create a virtual environment & activate it**

```
C:\Users\print\Downloads\djgo-project>python -m venv venv
```

```
C:\Users\print\Downloads\djgo-project>venv\scripts\activate

(venv) C:\Users\print\Downloads\djgo-project>
```

- **Install Django**

```
(venv) C:\Users\print\Downloads\djgo-project>pip install django
Collecting django
  Using cached django-6.1.1-py3-none-any.whl.metadata (3.9 kB)
Collecting asgiref>=3.9.1 (from django)
  Using cached asgiref-3.12.1-py3-none-any.whl.metadata (9.4 kB)
Collecting sqlparse>=0.5.0 (from django)
  Using cached sqlparse-0.6.0-py3-none-any.whl.metadata (6.0 kB)
Collecting tzdata (from django)
  Using cached tzdata-2026.4-py2.py3-none-any.whl.metadata (1.4 kB)
Using cached django-6.1.1-py3-none-any.whl (8.4 MB)
Using cached asgiref-3.12.1-py3-none-any.whl (25 kB)
Using cached sqlparse-0.6.0-py3-none-any.whl (50 kB)
Using cached tzdata-2026.4-py2.py3-none-any.whl (347 kB)
Installing collected packages: tzdata, sqlparse, asgiref, django
Successfully installed asgiref-3.12.1 django-6.1.1 sqlparse-0.6.0 tzdata-2026.4

[notice] A new release of pip is available: 24.0 -> 26.2.1
[notice] To update, run: python.exe -m pip install --upgrade pip

(venv) C:\Users\print\Downloads\djgo-project>python manage.py runserver
Watching for file changes with StatReloader
Performing system checks...

System check identified no issues (0 silenced).

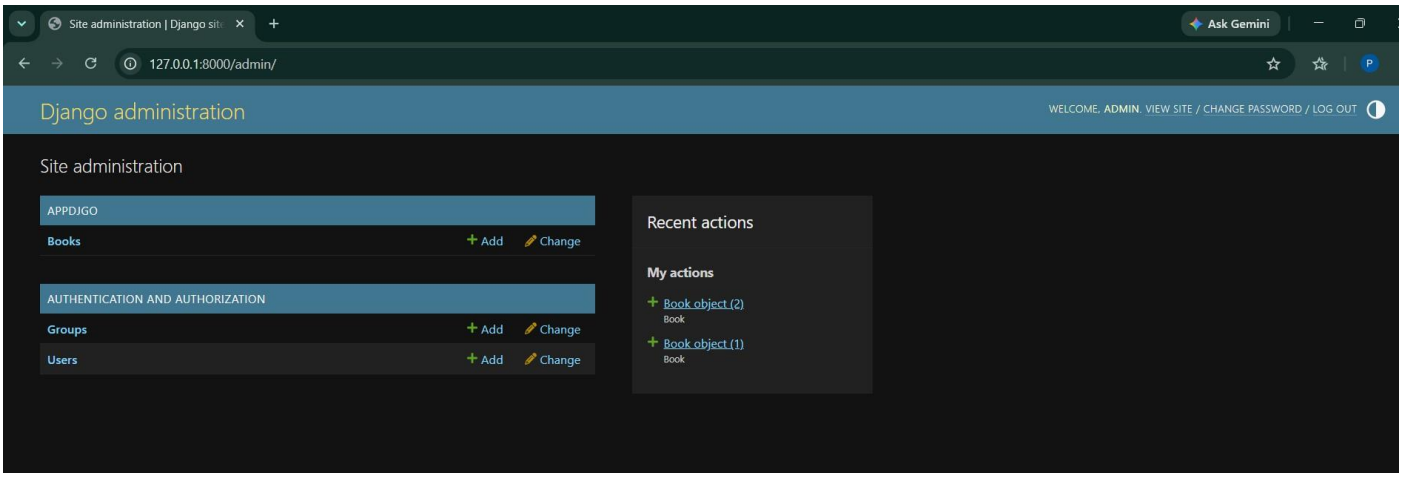
You have 18 unapplied migration(s). Your project may not work properly until you apply the migrations for app(s): admin, auth, contenttypes, sessions.
Run 'python manage.py migrate' to apply them.
October 01, 2026 - 19:39:09
Django version 6.1.1, using settings 'projectdjgo.settings'
Starting WSGI development server at http://127.0.0.1:8000/
Quit the server with CTRL-BREAK.
```

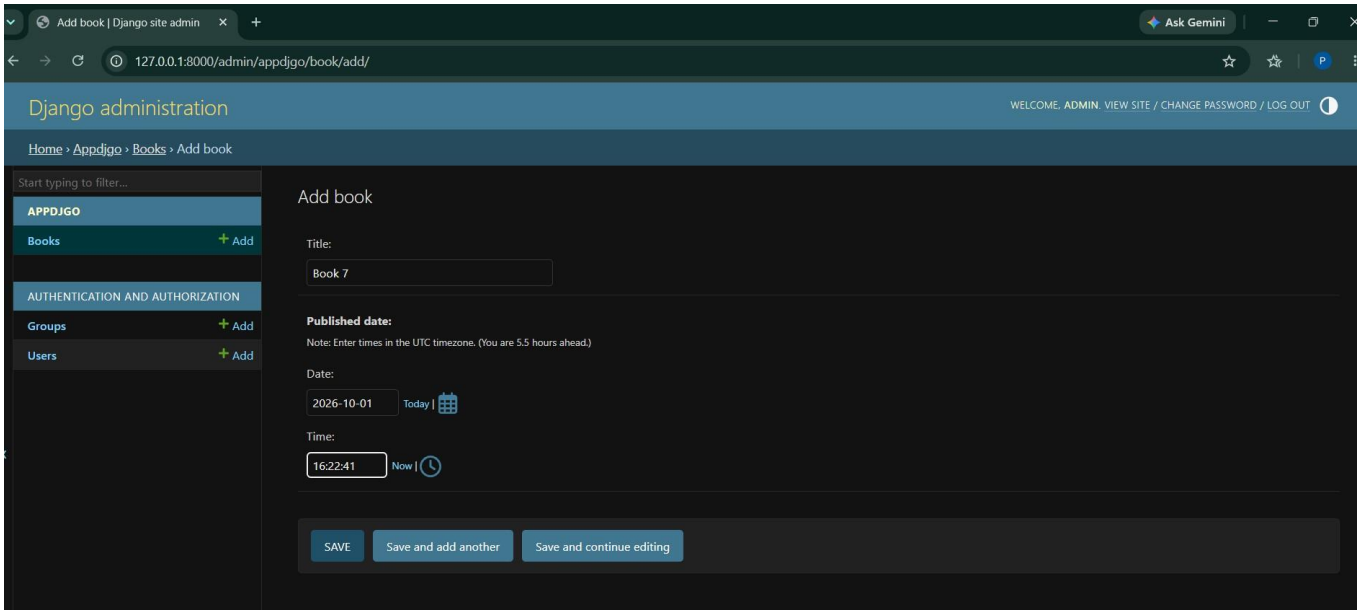
- **Create Project & App**

```
C:\Users\print\Downloads\djgo-project>django-admin startproject projectdjgo .

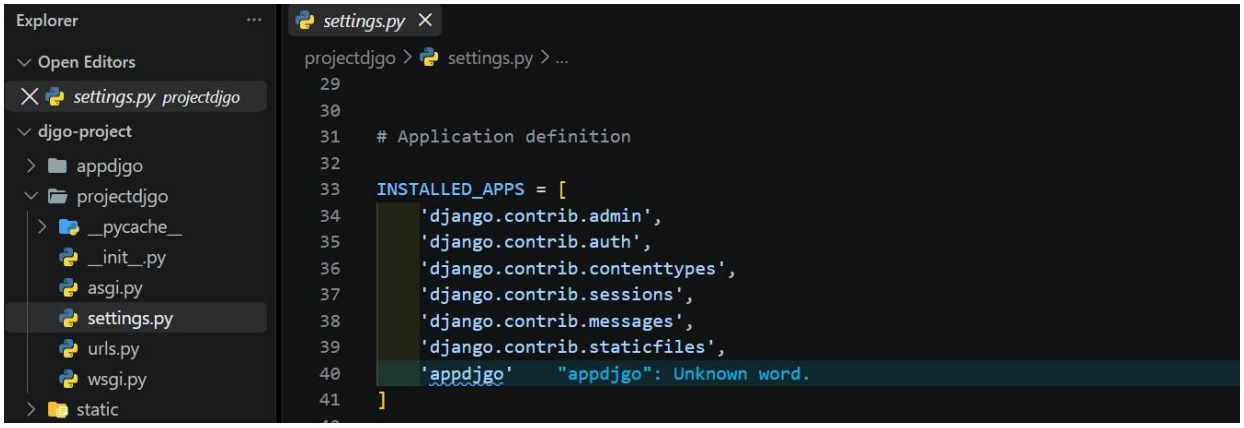
C:\Users\print\Downloads\djgo-project>django-admin startapp appdjgo
```

OUTPUT

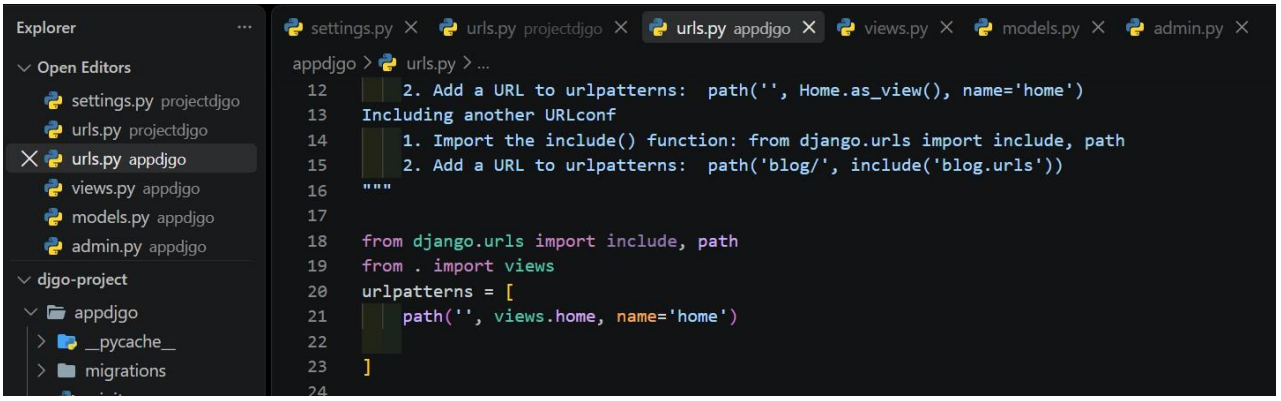

Site administration | Django site
Ask Gemini
127.0.0.1:8000/admin/
Django administration
WELCOME, ADMIN. VIEW SITE / CHANGE PASSWORD / LOG OUT
Site administration
APPDJGO
Books
Add
Change
AUTHENTICATION AND AUTHORIZATION
Groups
Add
Change
Users
Add
Change
Recent actions
My actions
Book object (2)
Book
Book object (1)
Book


Add book | Django site admin
Ask Gemini
127.0.0.1:8000/admin/appdjgo/book/add/
Django administration
WELCOME, ADMIN. VIEW SITE / CHANGE PASSWORD / LOG OUT
Home › Appdjgo › Books › Add book
Start typing to filter...
APPDJGO
Books
Add
AUTHENTICATION AND AUTHORIZATION
Groups
Add
Users
Add
Add book
Title:
Book 7
Published date:
Note: Enter times in the UTC timezone. (You are 5.5 hours ahead.)
Date:
2026-10-01
Today
Time:
16:22:41
Now
SAVE
Save and add another
Save and continue editing

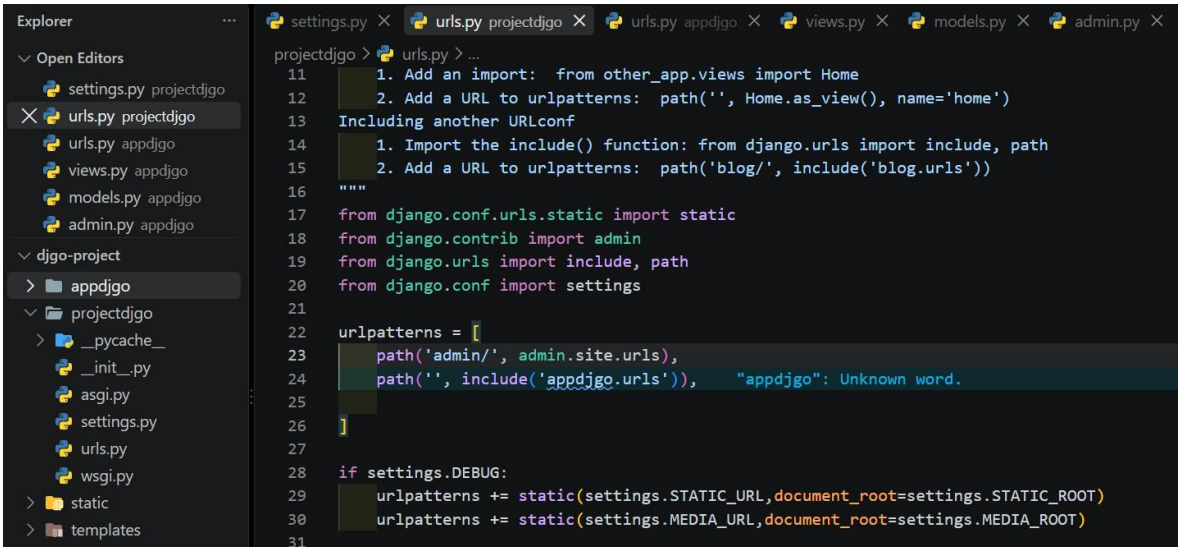
- **Add App name into installed apps**



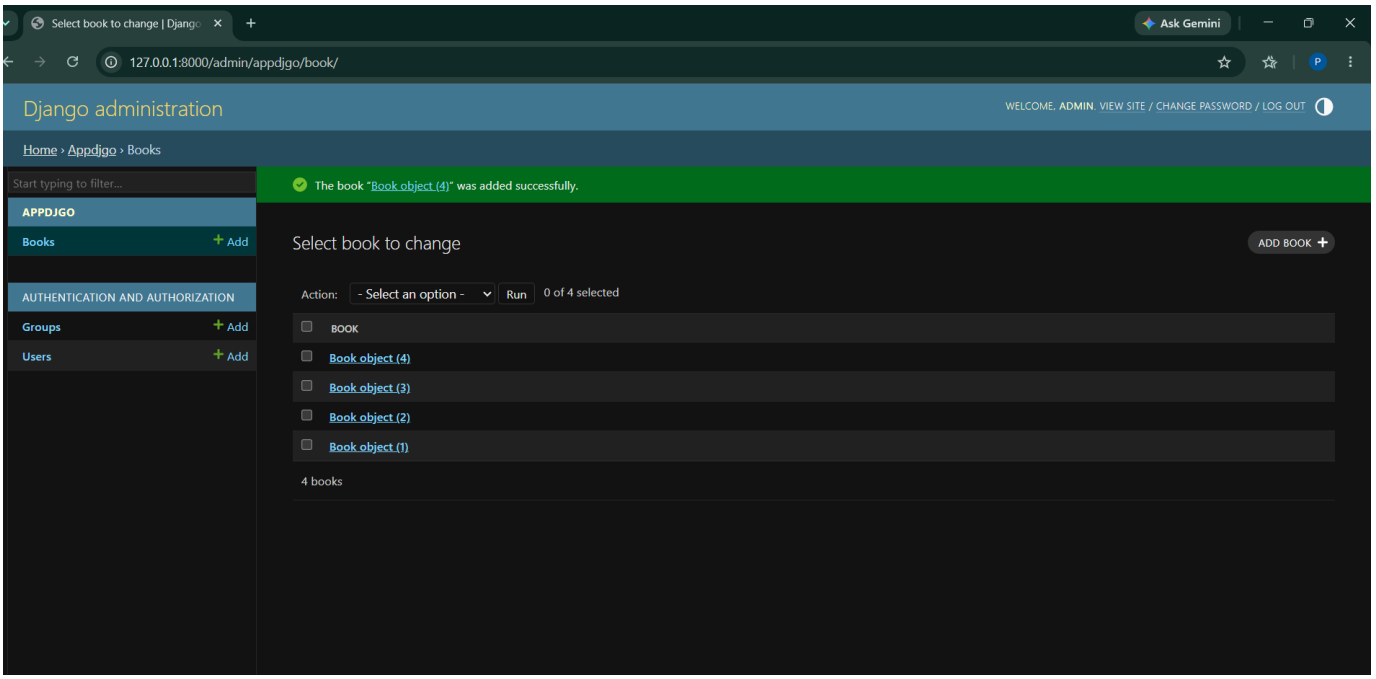
- **Give views and path in urls.py inside the app components**



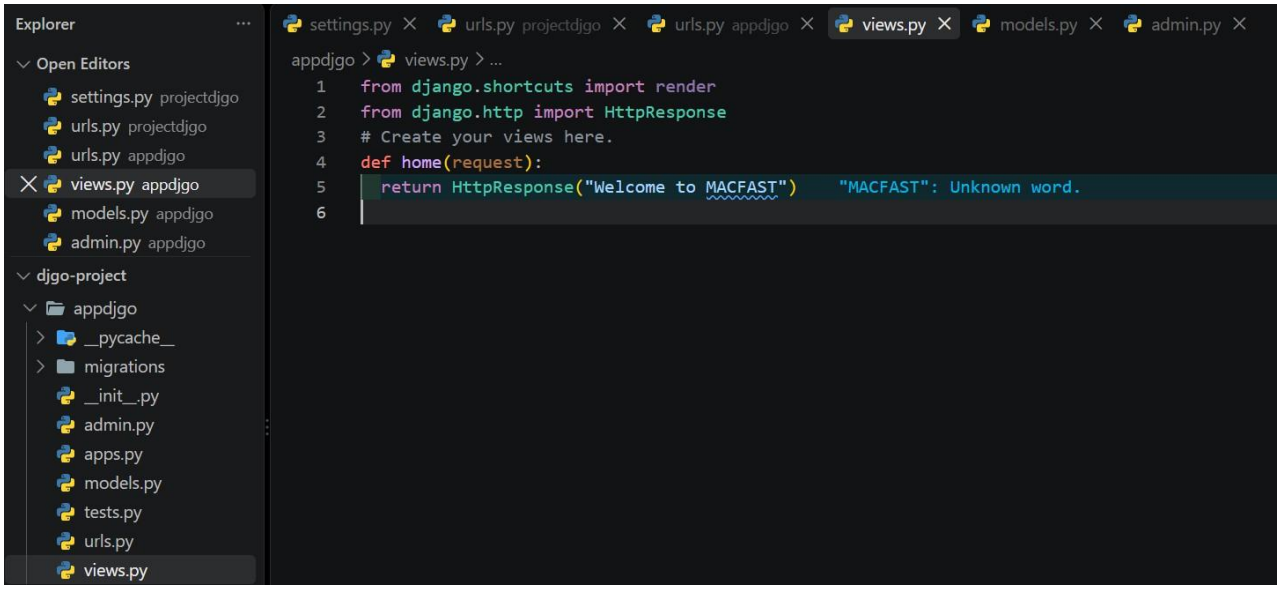
- **Give urls of static and media in urls.py in project components**



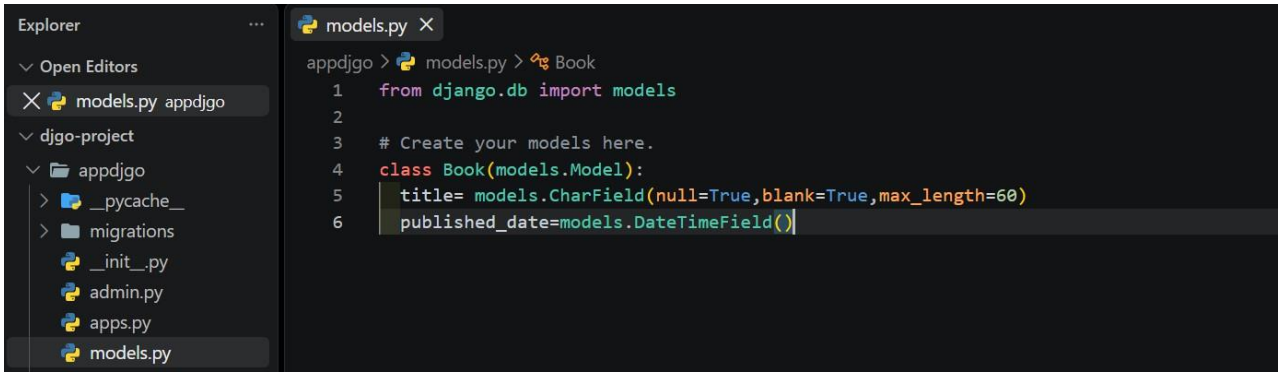
OUTPUT


Select book to change | Django
Ask Gemini
127.0.0.1:8000/admin/appdjgo/book/
Django administration
WELCOME, ADMIN. VIEW SITE / CHANGE PASSWORD / LOG OUT
Home › Appdjgo › Books
Start typing to filter...
APPDJGO
Books + Add
AUTHENTICATION AND AUTHORIZATION
Groups + Add
Users + Add
The book "Book object (4)" was added successfully.
Select book to change
ADD BOOK +
Action: - Select an option - Run 0 of 4 selected
BOOK
Book object (4)
Book object (3)
Book object (2)
Book object (1)
4 books

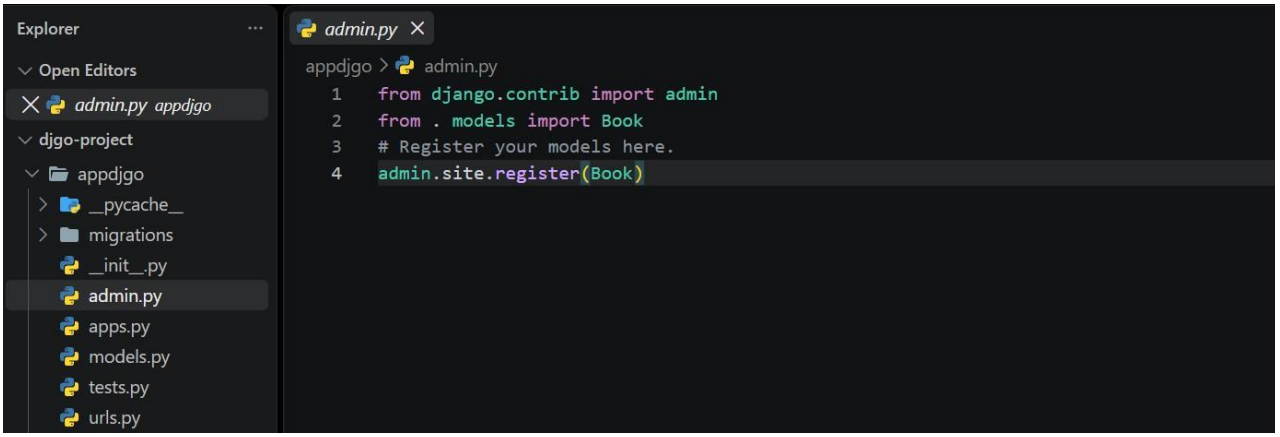
- **Create function in views.py inside app component**



- **Create a Database in models.py**



- **Register Database into admin.py**



- **Apply migrate into app component**

```
(venv) C:\Users\print\Downloads\djgo-project>python manage.py migrate
Operations to perform:
  Apply all migrations: admin, auth, contenttypes, sessions
Running migrations:
  Applying contenttypes.0001_initial... OK
  Applying auth.0001_initial... OK
  Applying admin.0001_initial... OK
  Applying admin.0002_logentry_remove_auto_add... OK
  Applying admin.0003_logentry_add_action_flag_choices... OK
  Applying contenttypes.0002_remove_content_type_name... OK
  Applying auth.0002_alter_permission_name_max_length... OK
  Applying auth.0003_alter_user_email_max_length... OK
  Applying auth.0004_alter_user_username_opts... OK
  Applying auth.0005_alter_user_last_login_null... OK
  Applying auth.0006_require_contenttypes_0002... OK
  Applying auth.0007_alter_validators_add_error_messages... OK
  Applying auth.0008_alter_user_username_max_length... OK
  Applying auth.0009_alter_user_last_name_max_length... OK
  Applying auth.0010_alter_group_name_max_length... OK
  Applying auth.0011_update_proxy_permissions... OK
  Applying auth.0012_alter_user_first_name_max_length... OK
  Applying sessions.0001_initial... OK
```

```
(venv) C:\Users\print\Downloads\djgo-project>python manage.py makemigrations
Migrations for 'appdjgo':
  appdjgo\migrations\0001_initial.py
    + Create model Book

(venv) C:\Users\print\Downloads\djgo-project>python manage.py migrate
Operations to perform:
  Apply all migrations: admin, appdjgo, auth, contenttypes, sessions
Running migrations:
  Applying appdjgo.0001_initial... OK
```

- **Create a superuser and give username, email, password**

```
(venv) C:\Users\print\Downloads\djgo-project>python manage.py createsuperuser
Username (leave blank to use 'print'): admin
Email address: admin10@gmail.com
Password:
Password (again):
This password is too short. It must contain at least 8 characters.
This password is too common.
This password is entirely numeric.
Bypass password validation and create user anyway? [y/N]: y
Superuser created successfully.
```

- **Run the project**

```
(venv) C:\Users\print\Downloads\djgo-project>python manage.py runserver
```